geog604 project

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```
# libraries
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(nnet)
```

```
## Warning: package 'nnet' was built under R version 4.5.2
```

```
library(ggplot2)
library(rpart.plot)
```

```
## Warning: package 'rpart.plot' was built under R version 4.5.2
```

```
## Loading required package: rpart
```

```
## Warning: package 'rpart' was built under R version 4.5.2
```

```
library(mapindia)
```

```
## Warning: package 'mapindia' was built under R version 4.5.2
```

```
library(sf)
```

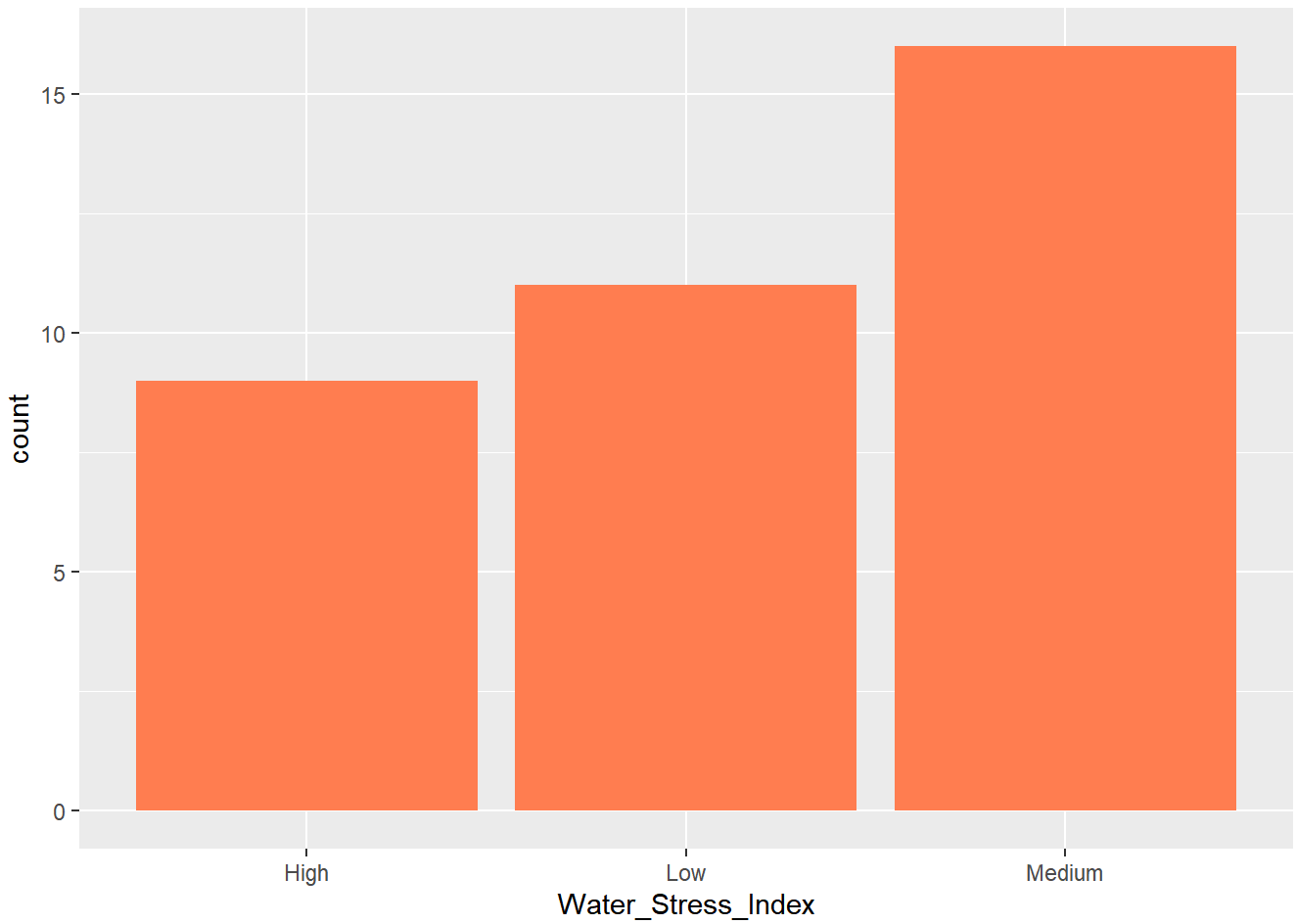
```
## Linking to GEOS 3.13.1, GDAL 3.11.0, PROJ 9.6.0; sf_use_s2() is TRUE
```

```
library(tmap)
```

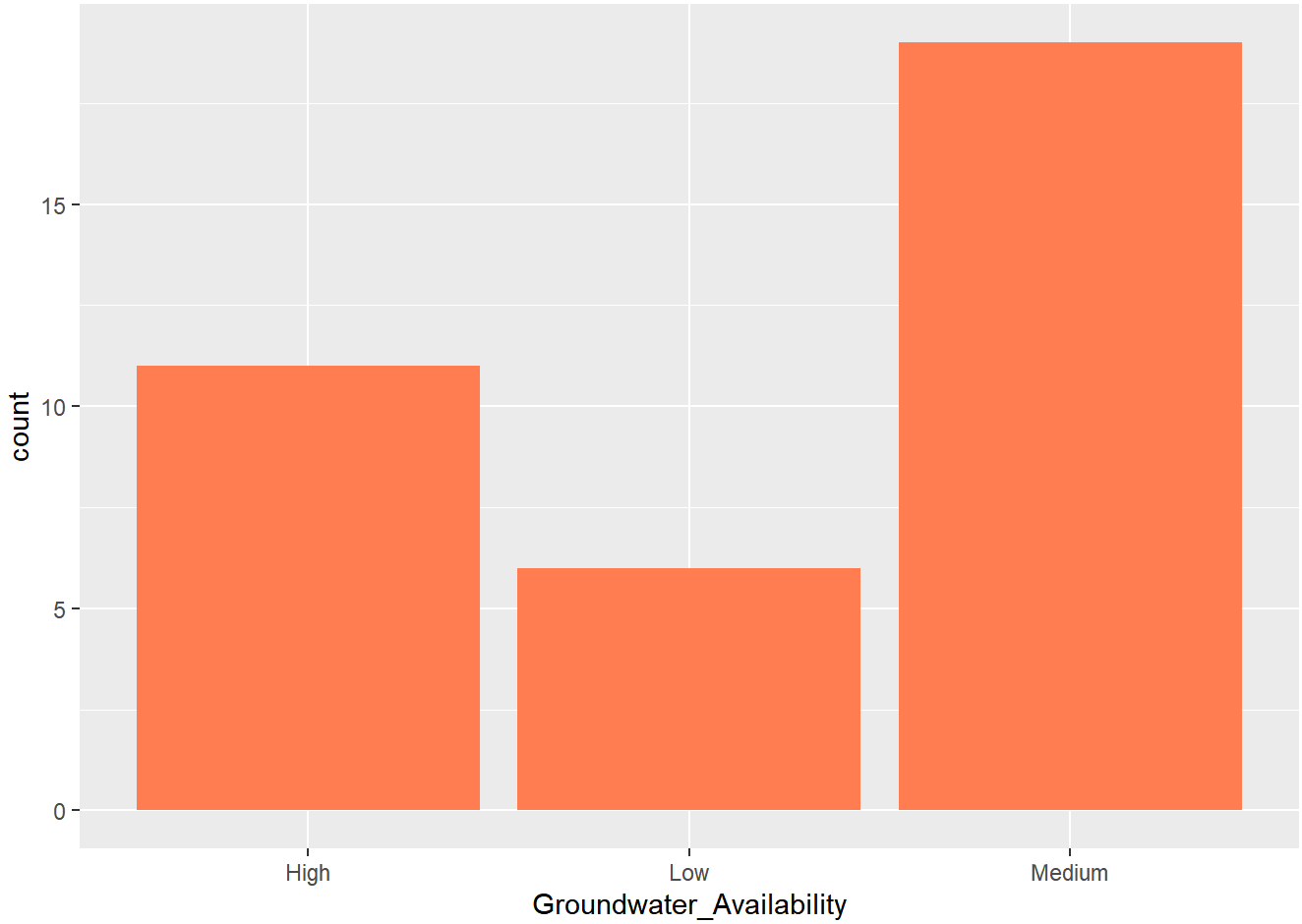
```
india = read.csv("C:/Users/theresa/Desktop/geog604/Agro_India_States.csv") # downloaded from kag
gle
```

Exploratory Data Analysis

```
ggplot(india) + geom_bar(aes(Water_Stress_Index), fill = 'coral')
```

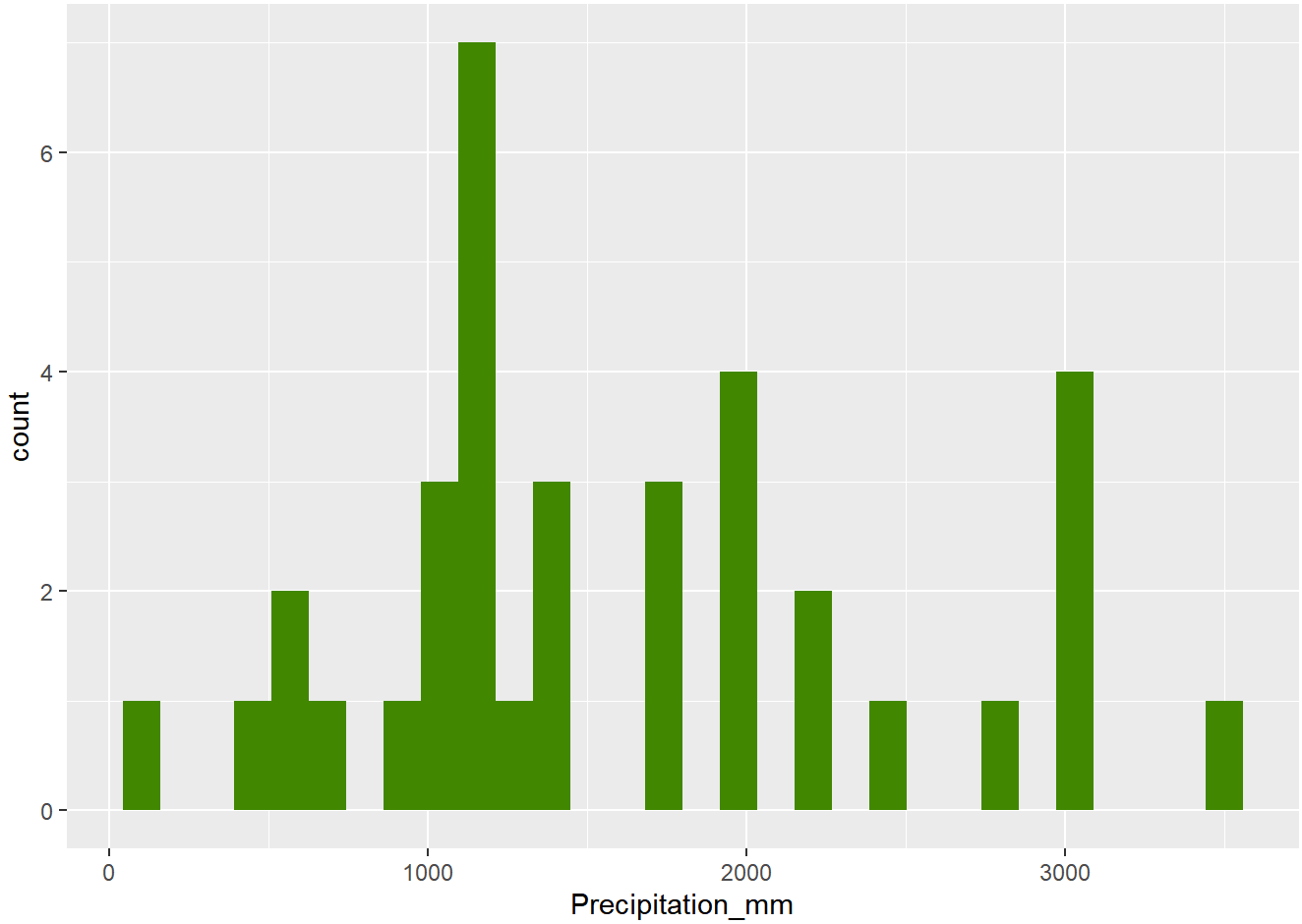



```
ggplot(india) + geom_bar(aes(Groundwater_Availability), fill = 'coral')
```

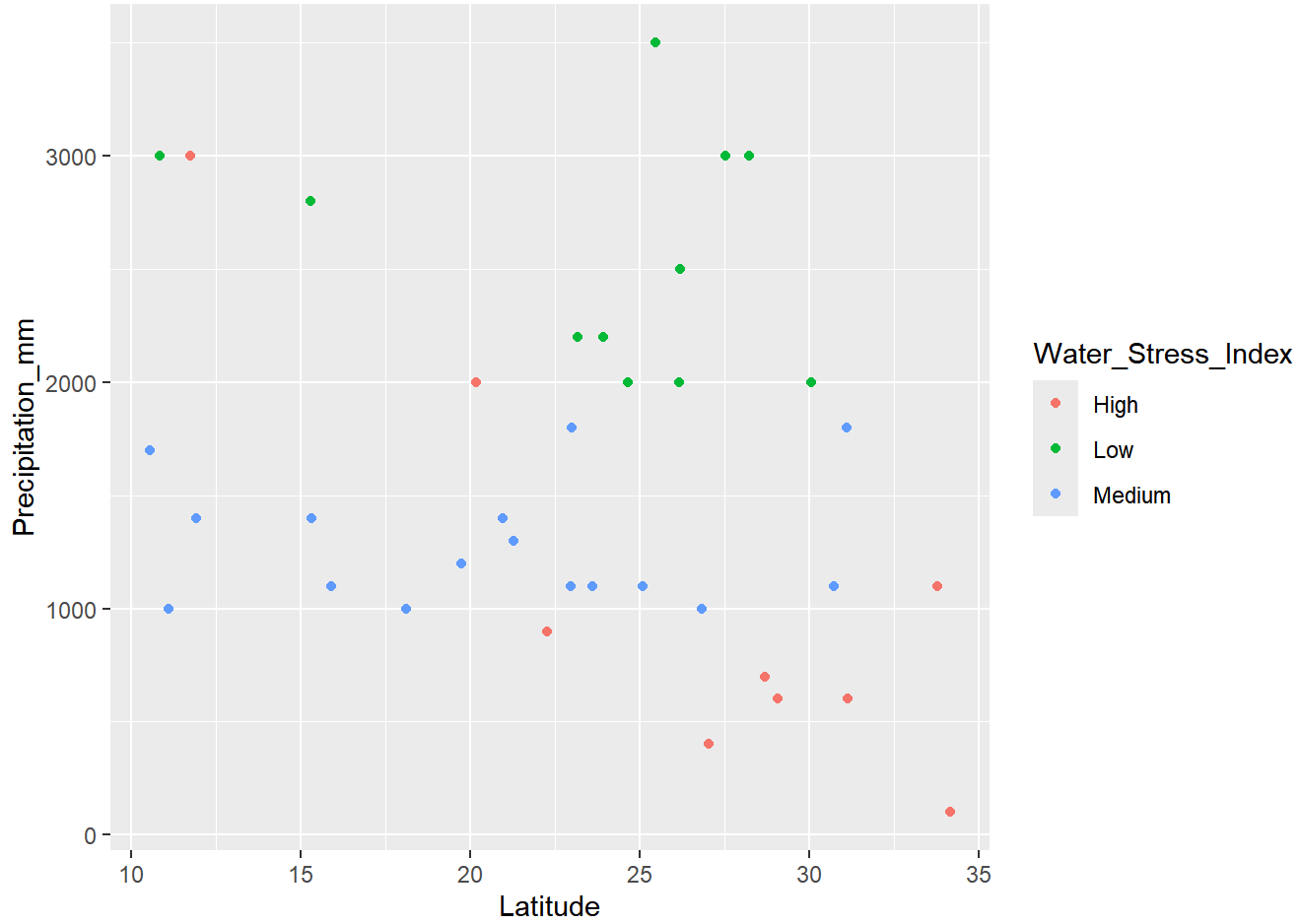



```
ggplot(india) + geom_histogram(aes(Precipitation_mm), fill = 'chartreuse4')
```

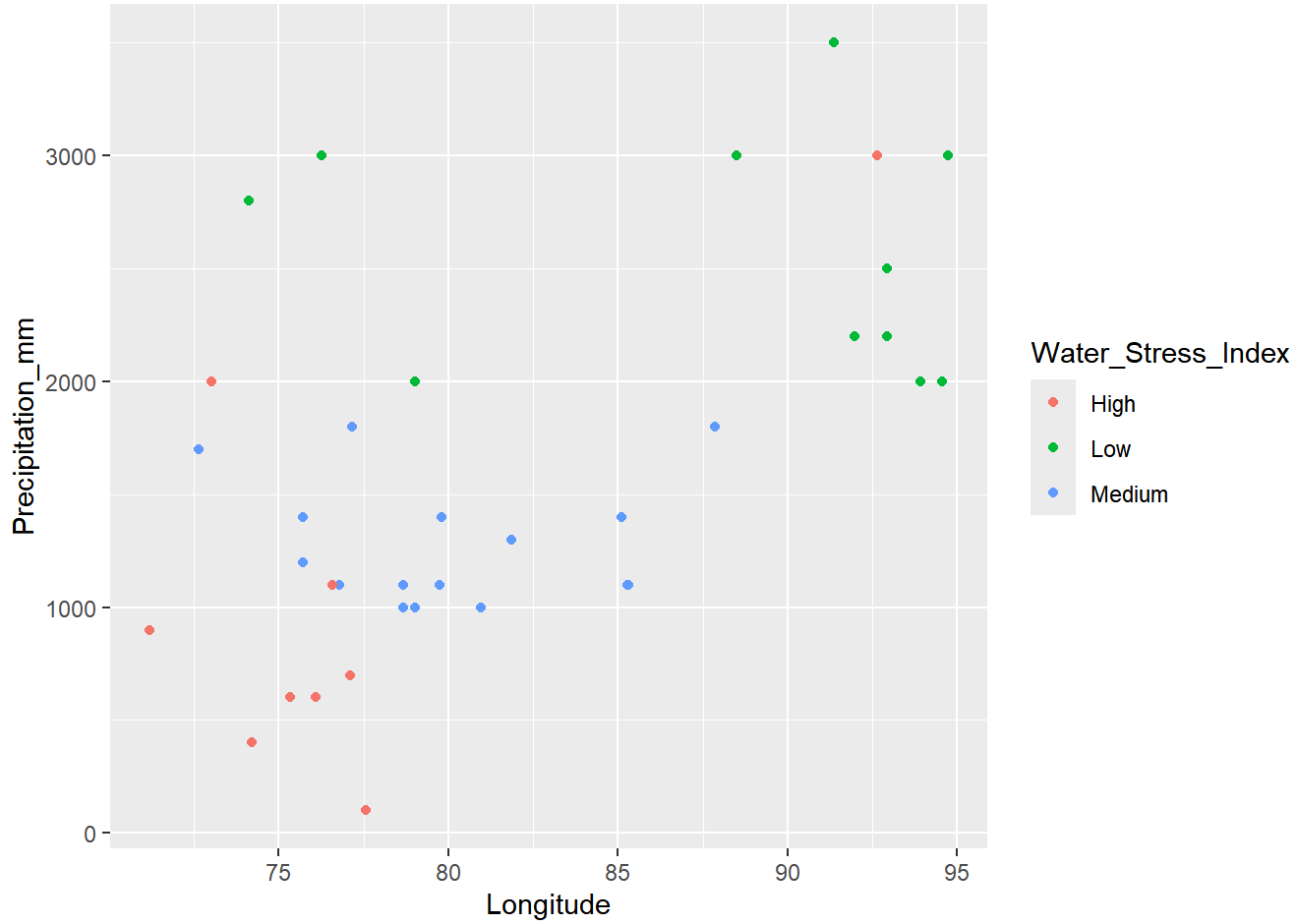
```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
ggplot(india) + geom_point(aes(x = Latitude, y = Precipitation_mm, col = Water_Stress_Index))
```




```
ggplot(india) + geom_point(aes(x = Longitude, y = Precipitation_mm, col = Water_Stress_Index))
```



Predicting water stress index

splitting the data 1:1

```
set.seed(10)
train_size = sample(nrow(india), size = nrow(india)/2)
india_train = india[train_size,]
india_test = india[-train_size,]
```

```
accuracy = function(x) {
  p = predict(x, india_test)
  t = table(p, india_test$Water_Stress_Index)
  acc = sum(diag(t) / sum(t))

  print(t)
  print(acc)
}
```

fitting the training data to model

```
# with all numerical predictors
fit = multinom(Water_Stress_Index ~ Precipitation_mm + Avg_Summer_Temp_C + Avg_Winter_Temp_C + L
ongitude + Latitude, data = india_train)
```

```
## # weights:  21 (12 variable)
## initial  value 19.775021
## iter  10 value 4.788487
## iter  20 value 0.050983
## iter  30 value 0.000329
## iter  40 value 0.000155
## iter  50 value 0.000138
## iter  60 value 0.000136
## iter  70 value 0.000124
## final  value 0.000086
## converged
```

```
summary(fit)
```

```
## Call:
## multinom(formula = Water_Stress_Index ~ Precipitation_mm + Avg_Summer_Temp_C +
##     Avg_Winter_Temp_C + Longitude + Latitude, data = india_train)
##
## Coefficients:
##        (Intercept) Precipitation_mm Avg_Summer_Temp_C Avg_Winter_Temp_C
## Low      -22.79983       0.01094345         -8.623565          1.364874
## Medium    24.34932      -0.12771942         -5.840778        -11.761020
##        Longitude   Latitude
## Low     4.654231  -4.580149
## Medium 10.389371 -11.765768
##
## Std. Errors:
##        (Intercept) Precipitation_mm Avg_Summer_Temp_C Avg_Winter_Temp_C
## Low       19.03969        2.1358102          83.21493          74.49130
## Medium    18.90368        0.7342126          75.55513          86.25034
##        Longitude Latitude
## Low     30.42632 9.232536
## Medium  12.38850 6.841306
##
## Residual Deviance: 0.0001714153
## AIC: 24.00017
```

```
accuracy(fit)
```

```
##
## p        High Low Medium
##   High      2   0      1
##   Low       0   5      0
##   Medium    3   1      6
## [1] 0.7222222
```

```
# with all numerical predictors - Longitude
fit = multinom(Water_Stress_Index ~ Precipitation_mm + Avg_Summer_Temp_C + Avg_Winter_Temp_C +
Latitude, data = india_train)
```

```
## # weights:  18 (10 variable)
## initial  value 19.775021
## iter  10 value 9.362911
## iter  20 value 0.752846
## iter  30 value 0.058749
## iter  40 value 0.047039
## iter  50 value 0.031245
## iter  60 value 0.029457
## iter  70 value 0.028741
## iter  80 value 0.027367
## iter  90 value 0.023014
## iter 100 value 0.020832
## final  value 0.020832
## stopped after 100 iterations
```

```
accuracy(fit)
```

```
##
## p        High Low Medium
##   High      4   0      0
##   Low       1   6      0
##   Medium    0   0      7
## [1] 0.9444444
```

```
# with all numerical predictors - Latitude
fit = multinom(Water_Stress_Index ~ Precipitation_mm + Avg_Summer_Temp_C + Avg_Winter_Temp_C + L
ongitude, data = india_train)
```

```
## # weights:  18 (10 variable)
## initial  value 19.775021
## iter  10 value 7.089547
## iter  20 value 0.097051
## iter  30 value 0.010747
## iter  40 value 0.001756
## iter  50 value 0.001415
## iter  60 value 0.001015
## iter  70 value 0.000842
## iter  80 value 0.000673
## iter  90 value 0.000507
## iter 100 value 0.000439
## final  value 0.000439
## stopped after 100 iterations
```

```
accuracy(fit)
```

```
##
## p        High Low Medium
##   High      4   0      0
##   Low       1   6      0
##   Medium    0   0      7
## [1] 0.9444444
```

```
# with all numerical predictors - Longitude&Latitude
fit = multinom(Water_Stress_Index ~ Precipitation_mm + Avg_Summer_Temp_C + Avg_Winter_Temp_C, da
ta = india_train)
```

```
## # weights:  15 (8 variable)
## initial  value 19.775021
## iter  10 value 7.598565
## iter  20 value 0.338249
## iter  30 value 0.296258
## iter  40 value 0.127551
## iter  50 value 0.099273
## iter  60 value 0.098116
## iter  70 value 0.074934
## iter  80 value 0.055964
## iter  90 value 0.054914
## iter 100 value 0.053901
## final  value 0.053901
## stopped after 100 iterations
```

```
accuracy(fit)
```

```
##
## p        High Low Medium
##   High      4   0      0
##   Low       1   5      0
##   Medium    0   1      7
## [1] 0.8888889
```

```
# only avg temps
fit = multinom(Water_Stress_Index ~ Avg_Summer_Temp_C + Avg_Winter_Temp_C, data = india_train)
```

```
## # weights:  12 (6 variable)
## initial  value 19.775021
## iter  10 value 16.201768
## iter  20 value 14.044699
## iter  30 value 13.751704
## iter  40 value 13.734400
## iter  50 value 13.733900
## iter  60 value 13.726769
## iter  70 value 13.726587
## iter  80 value 13.726163
## iter  90 value 13.725972
## iter 100 value 13.725733
## final  value 13.725733
## stopped after 100 iterations
```

```
accuracy(fit)
```

```
##
## p        High Low Medium
##   High      2   0      0
##   Low       1   1      0
##   Medium    2   5      7
## [1] 0.5555556
```

```
# with precipitation + lat + long
fit = multinom(Water_Stress_Index ~ Precipitation_mm + Latitude + Latitude, data = india_train)
```

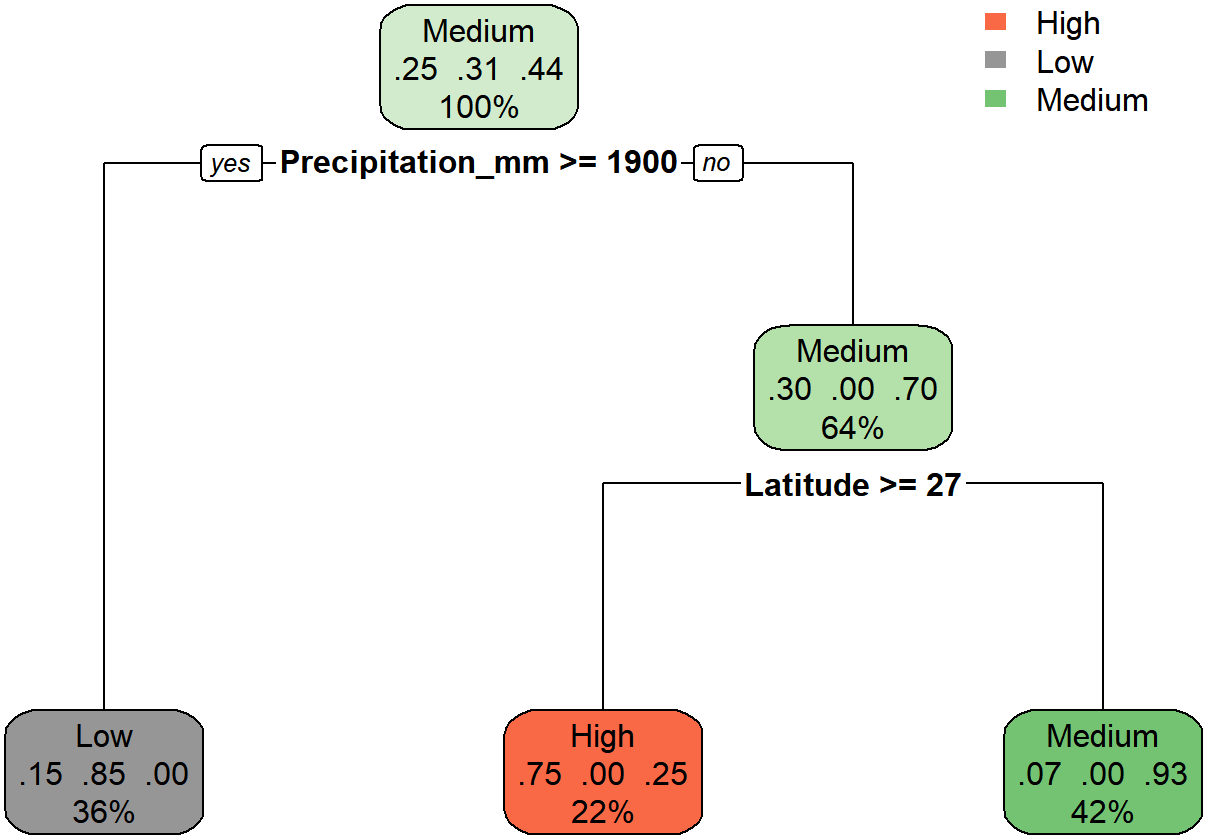
```
## # weights:  12 (6 variable)
## initial  value 19.775021
## iter  10 value 9.697851
## iter  20 value 9.300398
## iter  30 value 8.811287
## iter  40 value 8.596154
## iter  50 value 8.478509
## iter  60 value 8.386738
## iter  70 value 8.336252
## iter  80 value 8.274743
## iter  90 value 8.222921
## iter 100 value 8.094867
## final  value 8.094867
## stopped after 100 iterations
```

```
accuracy(fit)
```

```
##
## p        High Low Medium
##   High      4   0      1
##   Low       1   6      0
##   Medium    0   0      6
## [1] 0.8888889
```

tree diagram for water stress index

```
rtree = rpart(Water_Stress_Index ~ Precipitation_mm + Avg_Summer_Temp_C + Avg_Winter_Temp_C +  L
atitude + Longitude, data = india, cp = 0.008)
rpart.plot(rtree)
```



mapping for more exploratory analysis

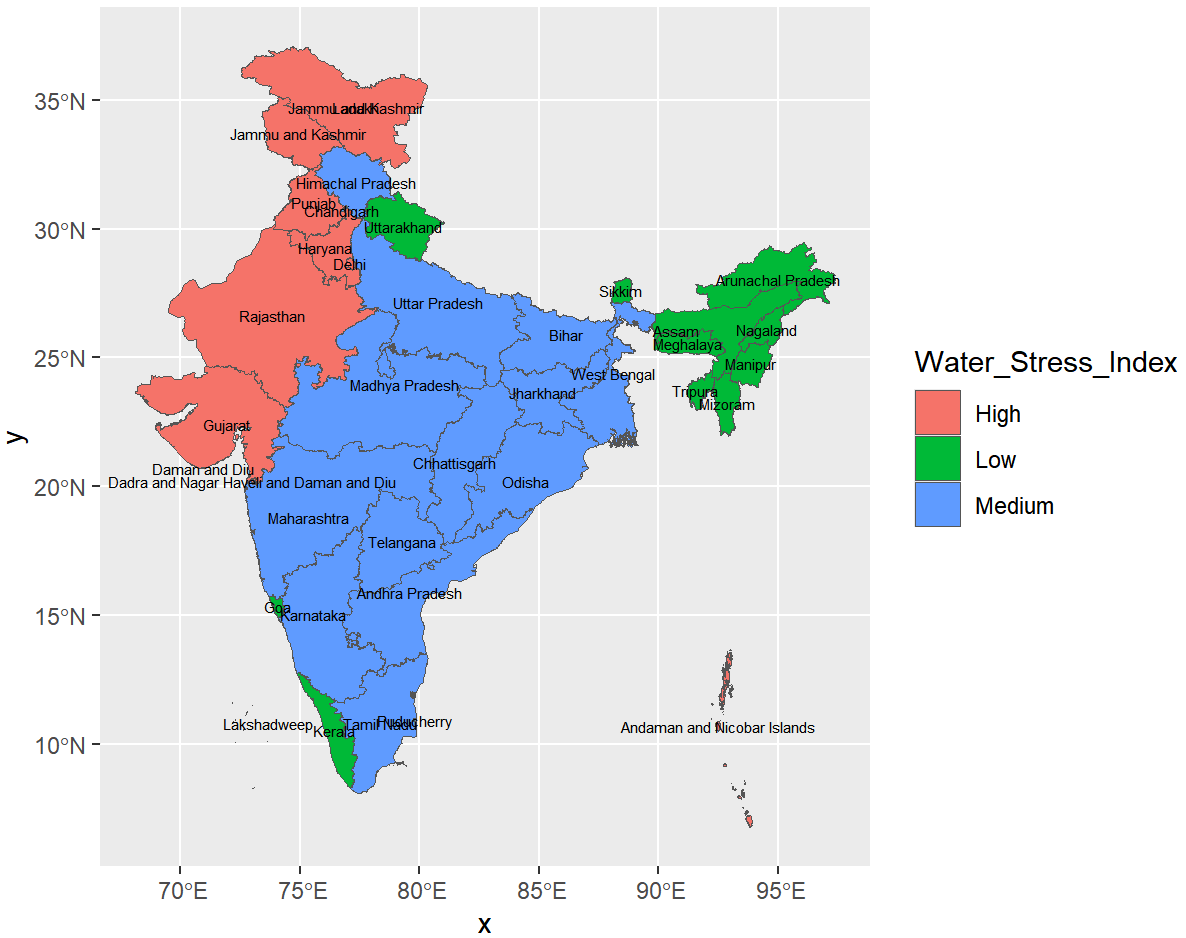
```
india_sf = st_as_sf(india, coords = c('Longitude', 'Latitude'), crs = 4326)
joined = st_join(map_india() ,india_sf, left = TRUE)

# fixing some NA value
joined$States[8] = 'Daman and Diu'
joined$Water_Stress_Index[8] = 'High'
joined$Precipitation_mm[8] = 2000
joined$Avg_Summer_Temp_C[8] = 32
joined$Avg_Winter_Temp_C[8] = 18
joined$Groundwater_Availability[8] = 'Medium'

joined$States[14] = 'Jammu and Kashmir'
joined$Water_Stress_Index[14] = 'High'
joined$Precipitation_mm[14] = 1100
joined$Avg_Summer_Temp_C[14] = 32
joined$Avg_Winter_Temp_C[14] = 12
joined$Groundwater_Availability[14] = 'Medium'
```

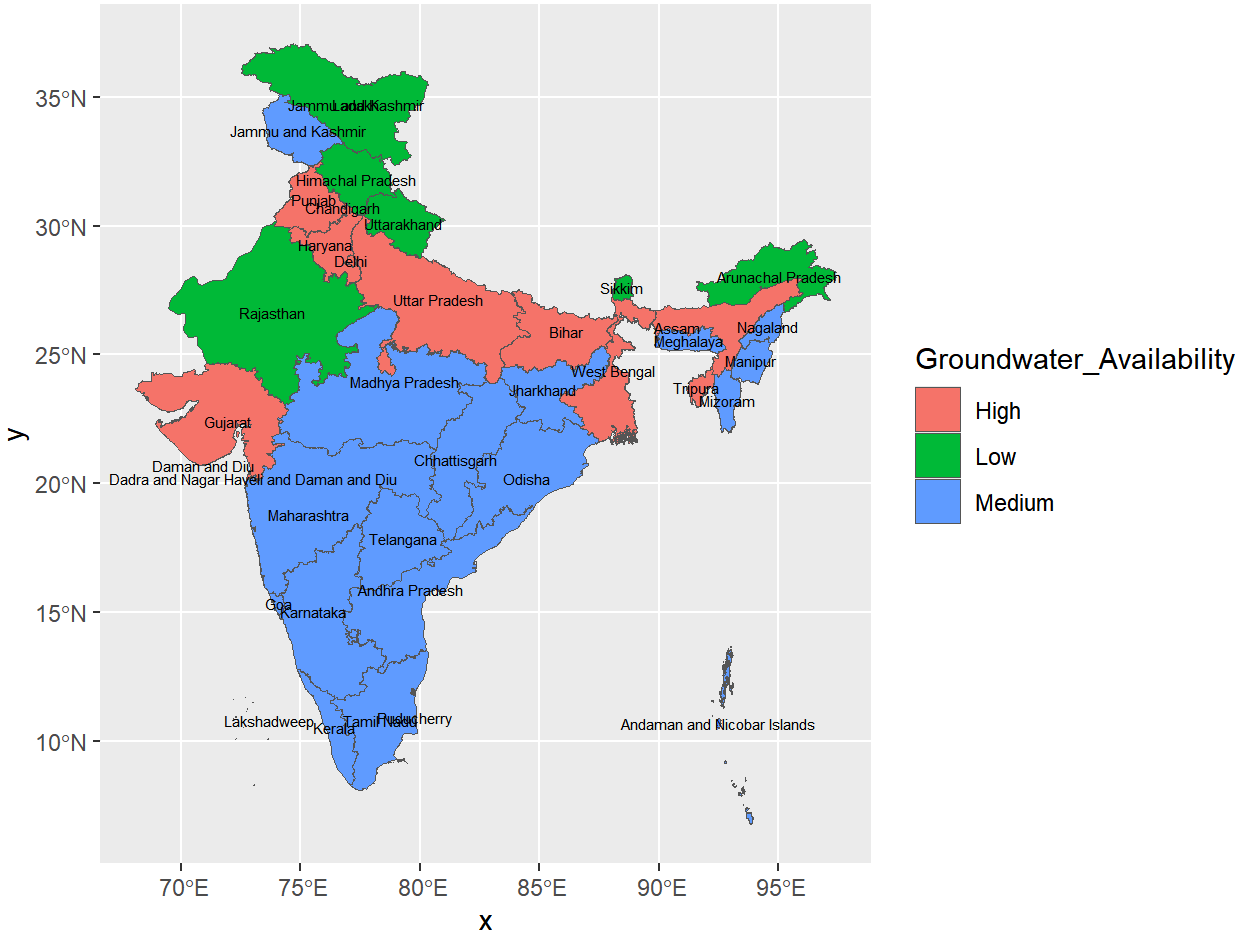
```
ggplot(joined) + geom_sf() + geom_sf(aes(fill= Water_Stress_Index)) + geom_sf_text(aes(label = S
tates), size = 2)
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not
## give correct results for longitude/latitude data
```



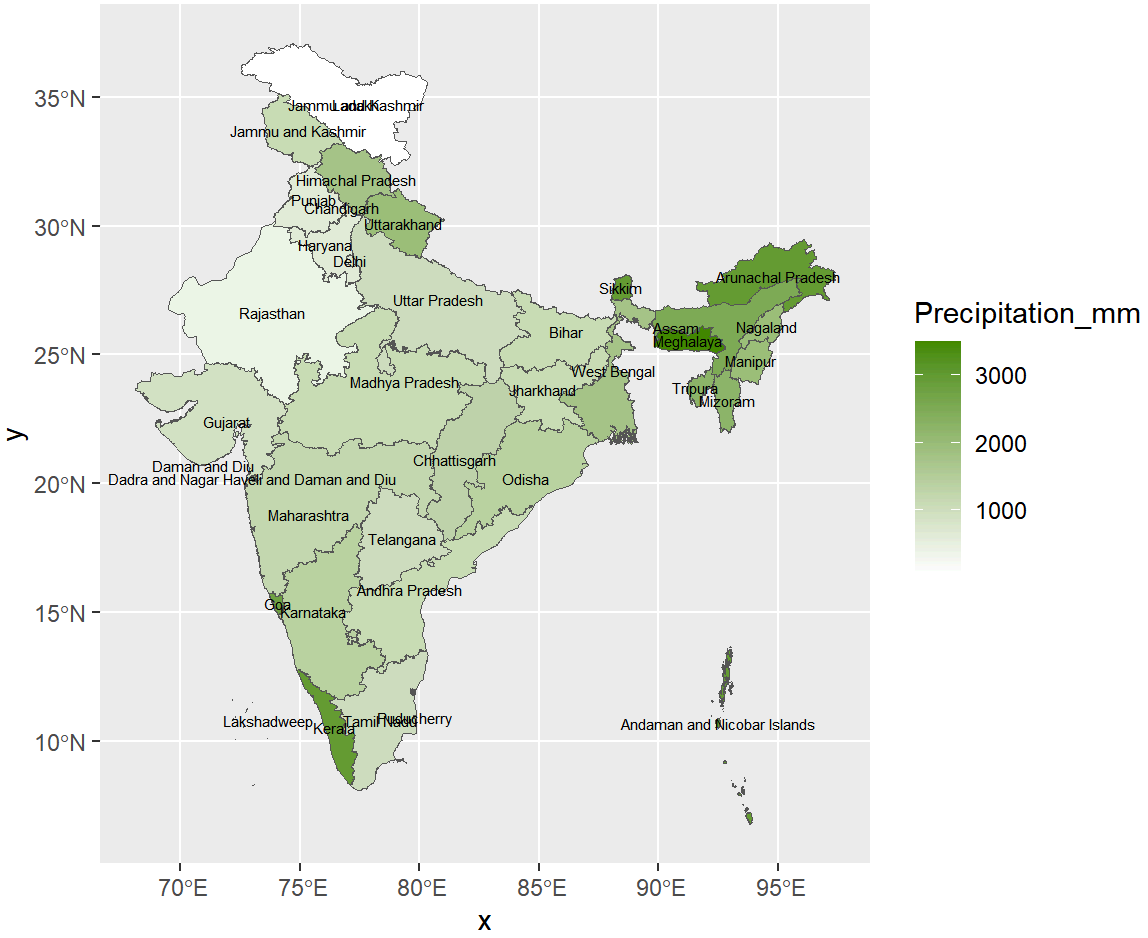

```
ggplot(joined) + geom_sf() + geom_sf(aes(fill= Groundwater_Availability)) + geom_sf_text(aes(lab
el = States), size = 2)
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not
## give correct results for longitude/latitude data
```



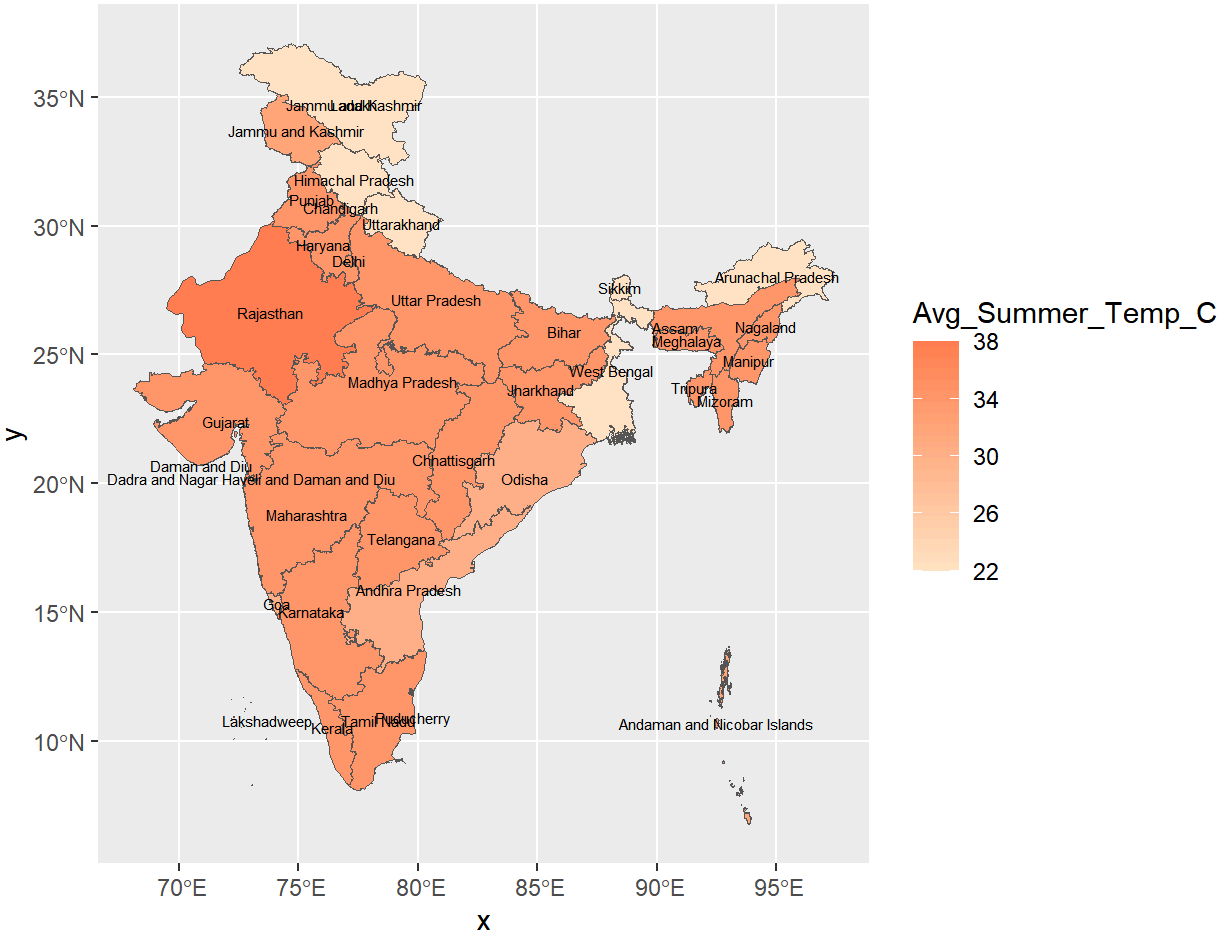

```
ggplot(joined) + geom_sf(aes(fill = Precipitation_mm)) + scale_fill_gradient(high = 'chartreuse
4', low = 'white') + geom_sf_text(aes(label = States), size = 2)
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not
## give correct results for longitude/latitude data
```



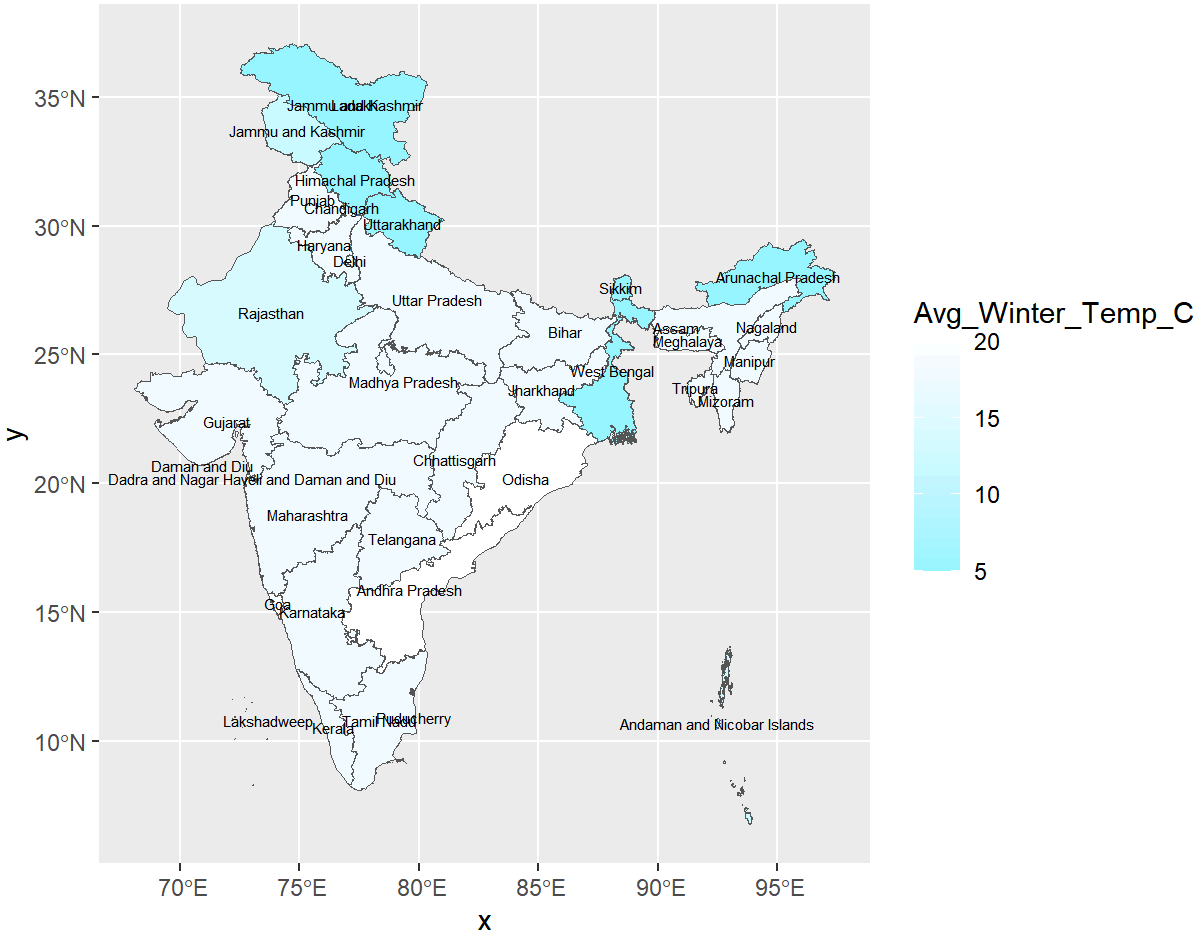

```
ggplot(joined) + geom_sf(aes(fill = Avg_Summer_Temp_C)) + scale_fill_gradient(high = 'coral', lo
w = 'bisque') + geom_sf_text(aes(label = States), size = 2)
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not
## give correct results for longitude/latitude data
```




```
ggplot(joined) + geom_sf(aes(fill = Avg_Winter_Temp_C)) + scale_fill_gradient(high = 'white', lo
w = 'cadetblue1') + geom_sf_text(aes(label = States), size = 2)
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not
## give correct results for longitude/latitude data
```


35°N
30°N
25°N
20°N
15°N
10°N
y
70°E
75°E
80°E
85°E
90°E
95°E
x
Jammu and Kashmir
Ladakh
Himachal Pradesh
Punjab
Chandigarh
Uttarakhand
Haryana
Delhi
Rajasthan
Uttar Pradesh
Sikkim
Arunachal Pradesh
Bihar
Assam
Meghalaya
Nagaland
Manipur
West Bengal
Tripura
Mizoram
Madhya Pradesh
Jharkhand
Gujarat
Daman and Diu
Dadra and Nagar Haveli and Daman and Diu
Chhattisgarh
Odisha
Maharashtra
Telangana
Andhra Pradesh
Goa
Karnataka
Lakshadweep
Kerala
Tamil Nadu
Puducherry
Andaman and Nicobar Islands
Avg_Winter_Temp_C
20
15
10
5